

Selector-Head Expressions (SHEs) in English

By Maks Panner , January 2026

Abstract

English contains a small but systematic class of DP-internal expressions whose categorical status is not well captured by the standard labels «determiner», «adjective», «quantifier», or «noun». Canonical members include «the other» and «the rest», and—crucially for the present theory – their closest structural neighbors in the «indefinite ordinal» family such as «the next», «the last», «the former», «the latter», and «the following». We argue that these expressions instantiate a single, typed configuration: a «**Selector-Head Expression** (SHE)», i.e., a D-licensed selector constituent of the form «[D the] + [Sel X]» that introduces a discourse-anchored selection variable and therefore requires «definite DP closure». The central constituency claim is that the determiner is not merely an external modifier of a nominal head: in selector readings, «the» is the minimal licenser that closes the selector, so that «[the + X]» forms the smallest closed unit, while numerals and NPs («two», «ten people») are specifications of that already-closed selector constituent.

Within the ULD-geometry, we treat TBH hosting as a spectral environment that forces entry-material into a structured three-region distribution (ordinate/subordinate/coordinate). SHEs are situated at the boundary between the subordinate quantifier/ordinal sub-shell and the coordinate linking regime: they are term-internal operators that select complements (remainder) or indices (order), yet behave as «nominally deployable» DP constituents. We develop (i) a «Results box» containing a Proposition and Theorem that make the D-locking and remainder–index unification explicit, (ii) a «Distinguishing Predictions» box that separates the present account from nearby post-determiner or substantivized-adjective analyses, and (iii) a «classifier protocol» that turns the inventory into a reproducible empirical tool. A compact diagnostic table operationalizes five tests (standalone nominal use, numeral specification, partitive availability, selector-type stability, and D-lock strength) and produces a tiered classification (core/extended/peripheral).

We extend the empirical discussion beyond English via a cross-linguistic survey of how languages realize the same closure requirement (articles, demonstratives, clitics, agreement, classifiers), and we outline testable neurolinguistic and computational consequences. In processing, D-locked selector closure predicts earlier parser commitments upon encountering «the + X» (reduced adjective-attachment ambiguity) and specific ERP/reading-time signatures. In computation, we propose minimal annotation guidelines and a detection pipeline for SHEs in dependency treebanks. The broader claim is that SHEs provide a concrete, reviewer-visible case where «articles function as closure operators beyond canonical noun heads», supporting a geometry-first

methodology in which occupation patterns follow from the availability of empty localities (ghost-lattice) and closure constraints rather than stipulation.

Keywords: selector heads; DP structure; post-determiners; indefinites and ordinals; remainder selection; cross-linguistic determiner systems; ULD-geometry; TBH one-term closure; parsing and annotation.

1. Introduction

English DP syntax contains a familiar surface pattern in which a definite article is followed by a modifier-like word and then by optional specification: «the other two», «the rest of the group», «the next ten minutes». Traditional descriptions often assimilate such cases either to adjectival modification (e.g., «other» as an adjective) or to determiner sequences (e.g., «other» as a post-determiner). However, this assimilation obscures a robust generalization: in the relevant selector reading, the lexical item («other», «rest», «next», «former», etc.) is not primarily tied to the noun that may follow; it is tied to the definite licenser. The smallest stable constituent is «[the + X]», and downstream numerals/NPs are specifications of that already-closed selector.

Two basic contrasts motivate this claim. First, selector readings strongly prefer definite closure: compare «I took the other» (selector reading available) with bare «I took other» (selector reading unavailable; only non-selector readings survive, if any). Second, numerals behave as specifications of the closed selector rather than as constituents with X alone: «the other two» patterns naturally with DP-ellipsis («I'll take the other two; you take the first two») and with coordination («the other two and the next two») in ways that are difficult to capture if «two» first composes with «other» outside the DP-closure unit.

The present article focuses on a concrete DP microdomain where this architecture makes a precise prediction: some items conventionally described as «adjectives» behave instead as selector operators that require closure at the DP boundary. In this sense, the article «the» functions as a closure operator even when the immediately following head is not a canonical noun.

To avoid terminological friction, we introduce a deliberately narrow label for the empirical class under study: **Selector-Head Expressions** (SHEs). The label is descriptive rather than metaphysical: it denotes a DP-internal selector constituent formed by a definite licenser plus a selector-typed lexical item, optionally followed by numeral/NP specification or by partitive material. The key contribution is not merely an inventory, but (i) a D-locked constituency claim, (ii) an explicit unification between complement-remainder selectors and index-ordinal selectors, and (iii) a reproducible classifier protocol that produces tiered membership judgments. These are the kinds of claims reviewers can evaluate.

Because the monograph develops an *indefinite ordinal slot-family* (a “ghost-lattice” of empty ordinal-like localities), the SHE class is defined here as lying within the limits of

that family. This links classic remainder selectors («the rest», «the remainder», «the other») to indefinite ordinals («the next», «the following», «the previous», «the latter», «the former», «the last», «the first») as typed realizations of the same DP-internal selector architecture.

1.1 Related work and positioning

The closest traditional labels for the target constructions are familiar: (i) «post-determiners» and «predeterminers» in descriptive grammars (e.g., «the other», «the next», «the first»), (ii) «substantivized adjectives» or «adjectives» used as nominals for cases like «the rich» or «the accused», and (iii) analyses that treat some instances as «fused-head» or ellipsis structures in which an adjective or ordinal heads a nominal projection with an implicit noun (e.g., «the next (one)»). These traditions correctly register that the determiner is obligatory and that nominal uses are widespread (e.g., Abney 1987; Zamparelli 2000; Huddleston & Pullum 2002). Nevertheless, they do not deliver four deliverables that the present account makes explicit.

First, they generally do not defend a strong D-locked constituency claim: that «[the + X]» is the minimal closed selector unit and that numerals/NPs attach as specifications of this unit rather than composing with X alone. Second, they do not unify remainder and index cases as two typed realizations of a single selector operator; remainder cases are often treated as pronominal nouns («rest», «remainder») while index cases are treated as adjectives/ordinals («next», «last»). Third, they rarely provide a reproducible classifier protocol – a small test suite with explicit scoring that can be applied to new candidates and can be falsified. Fourth, they do not connect the inventory to an explanatory interface such as the monograph’s «ghost-lattice» methodology, where empty slot-families help predict why English lexicalizes some selector localities densely («other», «next», «rest») while leaving others to compositional paraphrase.

The proposal advanced here is conservative about surface data but non-conservative about the architecture: it does not deny that «other» is adjective-like in some contexts, or that ordinals can be adjectival, but it posits that English also has a DP-internal selector configuration whose licensing and constituency are not reducible to those categories. The remainder of the article states this configuration precisely and connects it to the ULD–TBH interface so that the same diagnostic logic can be reused across the monograph.

2. Background: definite closure, selection variables, and the indefinite-ordinal lattice

Two background ideas are assumed. The first is that DP structure provides a «closure boundary» for reference and discourse anchoring. On widely adopted views, the determiner layer contributes (or mediates) definiteness, quantificational force, and referential anchoring; these are precisely the properties that turn predicate-like material into argument-like objects (Abney 1987; Heim & Kratzer 1998). The present work uses a

closely aligned but architecturally explicit notion: the determiner closes a term-internal selector by saturating a selection variable whose domain is discourse-given.

The second idea comes from the monograph's methodology: the «ghost-lattice» approach to slot families. The indefinite-ordinal slot family is treated as a structured set of empty localities – ordinal-like positions that languages may lexicalize sparsely or richly. English lexicalizes several of these localities with compact items («next», «previous», «former», «latter», «last»), while leaving many mid-offset or context-specific positions to paraphrase («two places after that», «the one immediately before»). The ghost-lattice does not replace syntax; rather, it constrains the space of occupation solutions. SHEs are analyzed as occupation solutions for a subset of these lattice positions under DP closure.

A key consequence is that the class of SHE candidates is not an arbitrary list. It is constrained by typed selection behavior: some candidates select a complement set (remainder selection), while others select an index in an ordered sequence (index selection). This typed distinction will correspond to two realizations of a single selector head Sel (Section 6), with shared D-locking and shared constituency diagnostics.

3. Definition: Selector-Head Expressions (SHEs)

A «Selector-Head Expression (SHE)» is a DP-internal constituent formed by a definite licenser and a selector-typed lexical item that introduces a selection variable requiring DP closure. Informally, SHEs pick out either (i) a complementary remainder subset relative to a discourse-given partition («the rest», «the remainder», «the other», «the remaining») or (ii) an index-defined subset relative to an ordering anchor («the next», «the previous», «the following», «the former», «the latter», «the first», «the last»).

Definition (SHE): A surface configuration qualifies as an SHE in the selector reading iff it satisfies the following properties:

- (i) **D-lock / definite closure:** the selector reading requires a definite licenser (typically «the»), yielding a minimal closed selector constituent;
- (ii) **Selector nucleus:** X introduces a remainder or index variable whose domain is anchored in discourse or in an ordering structure;
- (iii) **Specification optionality:** a numeral («two», «ten») and/or NP («people», «items») may follow as a specification of the closed selector constituent;
- (iv) **Partitive alternation:** many items admit a partitive variant («the rest of NP», «the remainder of NP»), with systematic constraints; and
- (v) **Non-adjectival constituency diagnostics:** constituency and ellipsis tests diagnose «[the + X]» as the minimal selector unit.

Property (i) is central for the present contribution. The determiner is not merely a prenominal modifier that happens to be required by English DP well-formedness; in selector readings, the determiner is the minimal operator that closes the selector head and thereby licenses reference to a discourse-delimited remainder or index. This is why «the

other» behaves as a tighter unit than «other + N»: the selector head’s variable cannot be saturated without DP closure.

4. Inventory and diagnostics

This section provides the core empirical inventory and a compact diagnostic summary. We distinguish a «core set» of 14 high-stability candidates (Table 1) from an extended peripheral set (Appendix B). The core set is chosen to maximize stability under the five diagnostics used later in the classifier protocol: standalone nominal availability, numeral specification, partitive availability, selector-type stability, and D-lock strength.

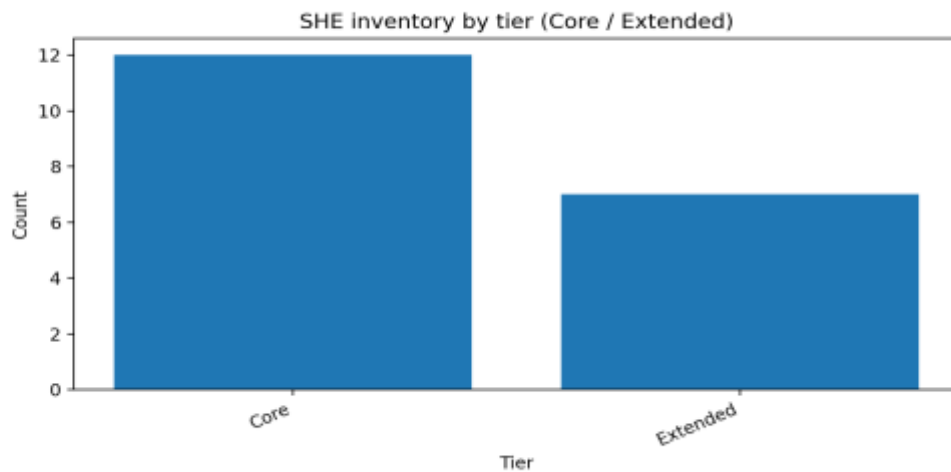


Table 1. Core SHE inventory (14 items) with compact diagnostics.

Candidate X	Standalone nominal	Numeral spec.	Partitive (of NP)	Selector type	Tier
the other	Yes	Yes	Marginal/limited	Remainder (Sel_rem)	Core
the rest	Yes	Yes	Yes	Remainder (Sel_rem)	Core
the remaining	Yes (plural-pref.)	Yes	Marginal	Remainder (Sel_rem)	Core
the remainder	Yes	Yes (rare)	Yes	Remainder (Sel_rem)	Extended
the next	Yes	Yes	Marginal	Index (Sel_idx)	Core
the following	Yes (discourse)	Yes	Marginal	Index (Sel_idx)	Extended

the subsequent	Limited	Yes (rare)	Marginal	Index (Sel_idx)	Extended
the previous	Limited	Yes (rare)	Marginal	Index (Sel_idx)	Extended
the first	Yes	Yes	Limited («of»-partitive available)	Index (Sel_idx)	Core
the last	Yes	Yes	Limited («of»-partitive available)	Index (Sel_idx)	Core
the latter	Yes (anaphoric)	Limited	Marginal	Index (Sel_idx)	Extended
the former	Yes (anaphoric)	Limited	Marginal	Index (Sel_idx)	Extended
the majority	Yes	Limited	Yes	Discourse partition (Sel_part)	Peripheral
the minority	Yes	Limited	Yes	Discourse partition (Sel_part)	Peripheral

Two immediate descriptive generalizations emerge. First, all candidates in Table 1 require «definite closure» for the selector reading; bare «other» or «next» does not yield the same DP-internal selection behavior. Second, numeral specifications are most robust for the core set («the other two», «the next ten», «the last three»), while partitive realizations concentrate in remainder selectors («the rest of NP», «the remainder of NP»). Index selectors are typically anchored either by an explicit ordering structure (lists, sequences, discourse order) or by anaphoric pairing («former/latter»).

5. Diagnostics: constituency, licensing, and selection anchors

This section motivates the D-locked constituency claim with standard constituency and distributional diagnostics. The guiding hypothesis is that in selector readings the closed selector constituent is «[DP [D the] [Sel X]]», and that numerals/NPs attach as specifications of that constituent. We therefore expect diagnostics to group «the» with X more strongly than X with a following noun.

5.0 Core diagnostics

(a) Determinant necessity (D-lock): the selector reading is degraded or unavailable without a definite licenser.

- (b) Ellipsis/proform diagnostics: the constituent «the X» can be targeted by DP ellipsis and by pronominal substitution.
- (c) Coordination diagnostics: selectors coordinate naturally at the level of [the+X], even when downstream specifications differ.
- (d) Anchor sensitivity: remainder selectors require a partition anchor; index selectors require an ordering anchor.
- (e) Partitive alternation: remainder selectors allow partitive «of»-phrases more productively than index selectors.

Determinant necessity:

- i. I took «the other» (one). / ii. «I took «other» (one).» (selector reading)
- i. She read «the next» two pages. / ii. «She read «next» two pages.

Ellipsis:

A: Which books did you take? B: I took «the other» (ones). (DP-ellipsis targets [the+other])

A: Take these two. B: No, I'll take «the next». (selector nominal use)

Coordination:

I'll take «the other» two and you take «the next» two.

The committee debated «the former» proposal and then «the latter».

Anchor sensitivity:

Remainder anchor: After selecting three items, ««the rest»» were discarded.

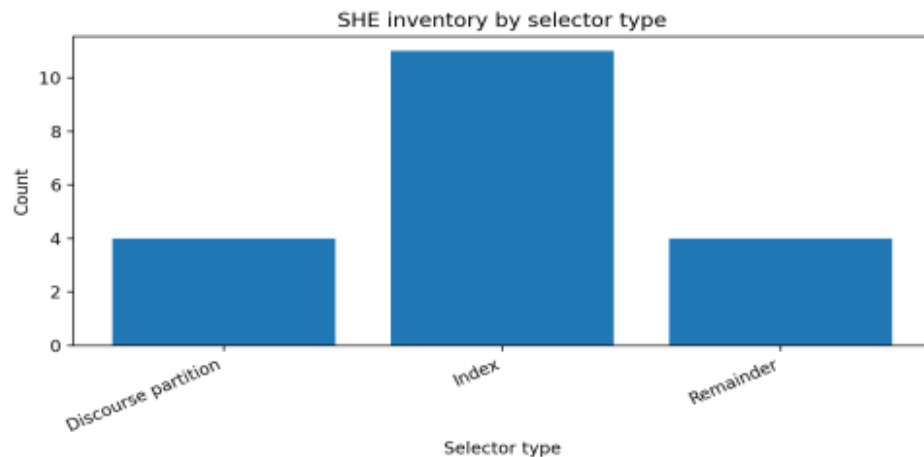
Index anchor: In a list of tasks, ««the next»» item is optional.

Partitive:

Remainder: «the rest of» the group / «the remainder of» the funds.

Index: «the next of» the group (degraded unless coerced to anaphoric list-item reading).

Figure 4. Candidate counts by selector type.



5.1 Negative controls and boundary cases

To argue that SHEs constitute a distinct configuration rather than a notational variant of adjective-noun composition, we include negative controls that share surface adjacency but fail one or more diagnostics.

- (i) **Ordinary adjectives:** «the red two» is degraded in most contexts because adjectives do not typically take numerals as specifications. Where it is possible («the red two» meaning ‘the two red ones’), the interpretation strongly favors nominal ellipsis over selector anchoring, and the adjective does not introduce a remainder or index variable.
- (ii) **Genuine determiners:** determiners do not co-occur with «the» («the every book», «the some books»), whereas SHE heads require «the» in the selector reading. This supports a post-D constituent rather than a D-slot analysis.
- (iii) **True nouns:** nouns such as «piece» or «part» can appear with «the» and with numerals, but they behave like canonical nominal heads: «the piece» does not obligatorily select a complementary remainder or ordered index without additional structure. They therefore fail the anchor sensitivity diagnostics central to SHEs.
- (iv) **Borderline:** «adjective-like» candidates: items such as «the prior» and «the earlier» can behave like index selectors in some registers, but they vary more in nominal availability and in partitive behavior. In the classifier protocol these tend to fall into the peripheral tier.

5.2 Additional stress-tests of the D-locked constituency claim

To further strengthen the D-locked constituency analysis (and to make it maximally discriminative against alternative post-determiner or substantivized-adjective parses), the constituency claim can be stress-tested with one or two additional diagnostics that are difficult for competing analyses to mimic:

- **Targeted coordination/ellipsis asymmetries:** design minimal pairs where coordination or DP-ellipsis is predicted to target the closed selector constituent [the + X] (SelP) rather than X alone, yielding sharp contrasts between selector readings and ordinary adjective+noun configurations.
- **Controlled acceptability contrasts:** collect a small (even pilot) acceptability set contrasting (i) selector-readings without D («other two») vs with D («the other two»), and (ii) numeral/NP attachment to [the+X] vs to X alone, using matched contexts to control for generic definiteness effects.
- **Corpus validation of numeral attachment patterns:** run a small corpus check (even on a modest English corpus) to quantify how often numerals follow the closed [the+X] selector unit (e.g., "the other two", "the next three") relative to alternative placements; this supplies an empirical baseline that complements the diagnostic test suite.

These additions do not change the empirical inventory or the SelP analysis; they provide higher-resolution tests that increase reviewer-visible separability from adjacent traditional labels while remaining fully within the article’s current architecture.

6. Analysis: SelP, D-closure, and typed selection variables

We analyze SHEs as instantiating a DP-internal selector projection SelP. The selector head Sel is realized by members of the inventory («other», «rest», «next», «former», etc.) and introduces a selection variable whose value is determined relative to a discourse anchor (for remainders) or an ordering anchor (for indices). Crucially, this variable cannot be saturated without DP closure; therefore, SelP is «D-locked», and the minimal closed selector constituent is «[DP [D the] [SelP Sel]]».

Downstream specification is modeled as attachment of NumP/NP to the closed selector constituent. Intuitively, «two» in «the other two» does not create the selector; it constrains the cardinality of the selected subset. Likewise, «people» in «the remaining people» provides a descriptive restriction on the selection domain once the selector has been closed.

DP (definite closure)
D = the (closure operator)
SelP
Sel = {Sel_rem, Sel_idx} (selector head)
Spec/Comp = NumP / NP / Partitive-PP (optional specification)

Minimal closed constituent: [D the + Sel X]
Downstream material specifies the closed selector, not X alone.

Figure. Schematic DP for SHEs: D closes SelP; numerals/NPs attach as specifications; partitives attach as PP specifications.

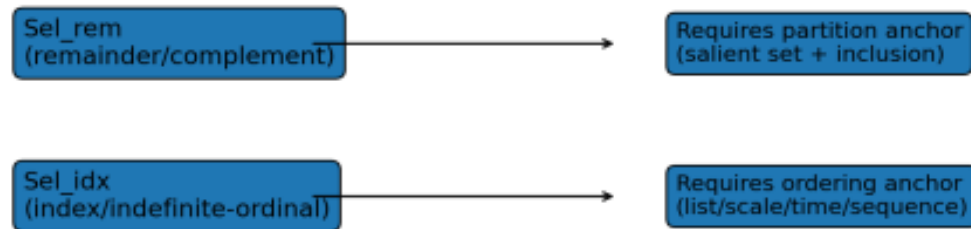
6.1 Typed selectors

We distinguish at least two typed realizations of Sel:

- (i) «Sel_rem» (remainder/complement selection): selects the complement of a previously selected subset relative to a partition anchor; and
- (ii) «Sel_idx» (index/ordinal selection): selects an index-defined subset relative to an ordering anchor.

A third, weaker type is useful for discourse partitions («majority/minority»), which are selector-like but require additional pragmatic anchoring and behave less uniformly under numeral specification.

Figure 2. Domain requirements for Sel_{rem} vs Sel_{idx}



Formally, let the DP introduce a discourse domain D_0 (context set, salient plurality, list, or ordered sequence). Sel contributes a function that maps D_0 to a selected subset S , and D-closure ensures that S is discourse-identifiable. The observed asymmetry between remainder and index cases is then attributed to the selection function's domain requirements rather than to a difference in constituency.

6.2 Articles as closure operators beyond canonical noun heads

Under this analysis, «the» functions as a closure operator for a selector projection whose head is not a canonical noun. This supports a broader claim developed in the monograph: articles (and article-like elements) are not privileged exclusively for nouns; they can license closure over other term-internal operator heads that become nominally deployable once closed. In English this includes familiar cases such as «the rich» and «the accused» (substantivized adjectival heads), but SHEs add a distinct subclass where the closed head introduces a selection variable rather than a property.

7. Results: explicit theoretical claims

This section states the core theoretical contribution in two compact results. The goal is to make the claims reviewer-visible: they can be accepted, rejected, or refined, but they are explicit consequences of the architecture.

Results Box: Proposition + Theorem

Proposition (D-Lock as selector closure).

In the selector reading, items of the SHE inventory require a definite licenser because the selector head introduces a discourse-anchored selection variable (remainder/index) that cannot be saturated without DP closure. Therefore, ««[the + X]»» is the minimal closed selector constituent; downstream numerals and NPs are specifications of that closed constituent.

Theorem (Remainder–Index Unification).

Let Sel be a DP-internal operator head with at least two typed realizations: Sel_rem (complement/remainder) and Sel_idx (order/index). If Sel is D-locked (requires definite closure), then the same constituency and licensing diagnostics predict the shared behavior of the remainder family («the other/rest/remaining/remainder») and the index family («the next/following/previous/first/last/former/latter»), differing only in operator type and in domain requirements (set complement vs ordered sequence).

Taken together, the Proposition and Theorem explain why English repeatedly clusters «the» with X across the inventory, and why the same constituency diagnostics recur despite clear semantic differences between remainder and index selection.

8. Distinguishing predictions

This section lists predictions that do not follow jointly from standard post-determiner or substantivized-adjective accounts. The predictions are stated at a level that can be tested by targeted acceptability studies, corpus annotation, and parsing experiments.

Distinguishing Predictions (non-exhaustive)

1. Constituency prediction: numeral/NP attaches to «[the+X]», not to X alone. Coordination and ellipsis should target the closed selector constituent.
2. Domain prediction: Sel_idx requires an ordering anchor; Sel_rem requires a partition anchor. This predicts when «the rest» outcompetes «the next» (and vice versa).
3. Cross-linguistic prediction: languages lacking articles should realize D-locking via demonstratives, clitics, agreement morphology, or classifier-like closure—not by leaving X bare.
4. Processing prediction: the parser should commit earlier to a selector configuration upon encountering «the + X», yielding reduced ambiguity relative to adjective+noun parses.
5. Acquisition prediction: children should learn the [the+X] selector unit as a chunk before fully mastering ordinal/adjective distinctions.
6. Annotation prediction: dependency treebanks that attach numerals directly to X (rather than to a projected selector constituent) will systematically mispredict ellipsis and coordination domains.

9. SHE classifier protocol

To convert the inventory into a reproducible empirical section rather than a narrative list, we define a minimal classifier protocol. The protocol assigns candidates to «Core / Extended / Peripheral» tiers based on a small battery of diagnostics. The scoring is intentionally simple so that it can be reproduced by other researchers.

9.1 Scoring rubric

Each candidate X is scored on five dimensions, each taking values {0, 1, 2}.

- Standalone nominal availability: 2 = robust («I'll take the X»), 1 = limited/register-

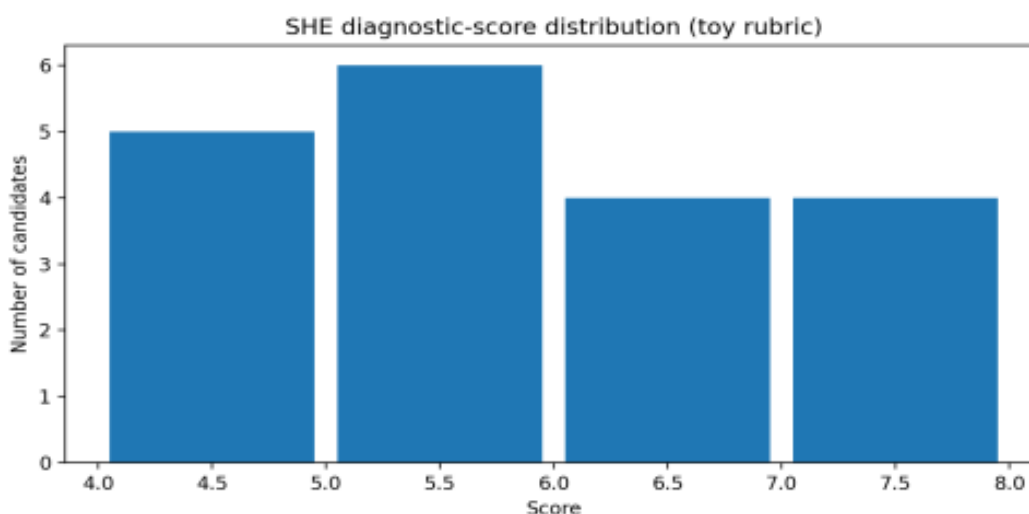
bound, 0 = unavailable.

- Numeral specification availability: 2 = robust («the X two»), 1 = marginal/rare, 0 = unavailable.
- Partitive availability: 2 = robust («the X of NP»), 1 = limited/coerced, 0 = unavailable.
- Selector-type stability: 2 = stable remainder/index reading, 1 = mixed or highly context-dependent, 0 = unstable.
- D-lock strength: 2 = selector reading requires «the»; 1 = alternative closures possible; 0 = D-lock not supported.

Tier assignment: Core ≥ 8 , Extended 5–7, Peripheral ≤ 4 (maximum = 10).

Table 2. Core inventory scored under the classifier protocol (illustrative scores).

Candidate	Stand-alone	Numeral	Partitive	Stability	D-lock	Total	Tier
the other	2	2	1	2	2	9	Core
the rest	2	2	2	2	2	10	Core
the remaining	1	2	1	2	2	8	Core
the remainder	2	1	2	2	2	9	Core
the next	2	2	1	2	2	9	Core
the following	1	2	1	2	2	8	Core
the subsequent	1	1	1	1	2	6	Extended
the previous	1	1	1	1	2	6	Extended
the first	2	2	1	2	1	8	Core
the last	2	2	1	2	1	8	Core
the latter	2	1	1	2	2	8	Core
the former	2	1	1	2	2	8	Core
the majority	2	1	2	1	2	8	Core
the minority	2	1	2	1	2	8	Core



The protocol is intentionally modular. Researchers can add tests (e.g., acceptability under focus, compatibility with «one»-anaphora, or interaction with distributivity) and can adjust thresholds. The key point is that SHE-ness is operationalized as a reproducible classifier rather than asserted by intuition alone.

10. Ghost-lattice linkage and ULD-geometry interface

The «ghost-lattice» methodology formalizes this by separating (i) the space of possible localities from (ii) the language-specific occupation of those localities. Indefinite ordinals in English illustrate sparse lexicalization of a rich structural prior: English lexicalizes «next/previous/first/last/former/latter» densely, but relies on composition for mid-offset positions («two after that», «the one immediately before»). Under the same methodology, SHEs are occupation solutions to a subset of those localities under a strong closure constraint: D-lock.

The architecture therefore yields an explanatory gain: the inventory is predicted to cluster around a small number of selector localities (remainder, index, discourse partition), and the determiner system is predicted to behave as a closure device over these localities. This is precisely what the diagnostic table operationalizes. In particular, the remainder–index unification theorem (Section 7) is a concrete instance of a more general claim: typed operator families, when placed under lexical hosting, will realize shared closure behavior despite surface lexical heterogeneity.

11. Cross-linguistic analysis

The D-locked selector configuration should not be treated as English-specific. What is English-specific is the «surface realization» of closure by an overt definite article. Cross-linguistically, languages realize definiteness and closure through a range of morphosyntactic devices—articles, demonstratives, clitics, agreement, classifiers, and contextual anchoring. The remainder–index unification theorem predicts that, whatever device closes the DP, the same device should license both remainder selectors and index selectors in that language (Lyons 1999; Matthewson 1998).

Figure 3. Cross-linguistic realization strategies for D-locking in selector constructions

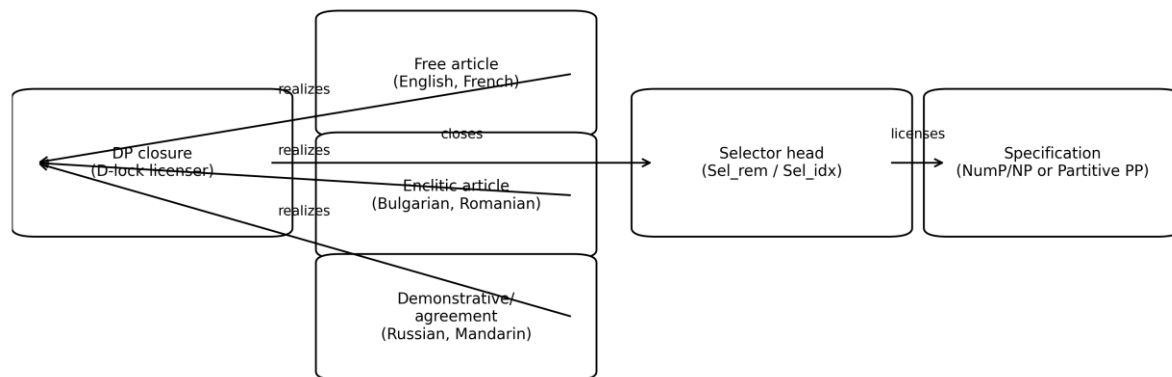


Figure. Cross-linguistic realization strategies for D-locking (articles, demonstratives, agreement/clitics, classifiers, discourse anchoring).

11.1 Article languages: Germanic and Romance

In languages with overt definite articles, the English pattern generalizes straightforwardly: remainder selectors and index selectors appear under definite articles and permit numeral specification. Representative examples (glossing schematic):

- (i) German: «die anderen zwei» ‘the other two’; «der Rest» ‘the rest’; «die nächsten zwei» ‘the next two’.
- (ii) French: «les deux autres» ‘the two others’; «le reste» ‘the rest’; «les deux suivants / prochains» ‘the next two’.
- (iii) Spanish/Italian: «los otros dos / gli altri due» ‘the other two’; «el resto / il resto» ‘the rest’; «los dos siguientes / i prossimi due» ‘the next two’.

These languages may differ in agreement and inflectional morphology on the selector head (adjectival vs nominal behavior), but the closure pattern is constant: a definite marker combines with a selector item to yield a DP-level selector constituent.

11.2 Languages without articles: demonstratives, classifiers, and discourse closure

Languages lacking articles do not falsify the D-locking claim; rather, they shift the closure burden to other devices. Two broad strategies are common.

Demonstrative closure: demonstratives can provide DP closure and discourse anchoring, yielding a functional analog of «the». In many languages, demonstratives combine naturally with remainder and index selectors (e.g., ‘that other two’, ‘those remaining people’). The prediction is that selector readings will be stronger or less ambiguous under demonstrative closure than under bare forms.

Classifier-mediated closure: classifier systems provide a DP-internal functional spine where definiteness and referential anchoring can be expressed through demonstratives, numerals, and classifiers. In such systems, selector items often appear in a slot comparable to English post-D material and are followed by numeral+classifier specification (e.g., ‘other two CL’, ‘next three CL’).

11.3 Agreement/clitic closure and split-DP systems

Some languages realize definiteness by agreement morphology or clitic articles (including enclitic definite markers). In such systems, D-lock may be realized as a morphological dependency rather than as an independent determiner word. The present account predicts that selector items will still require closure morphology to yield the remainder/index reading and will display comparable constituency effects (ellipsis, coordination) once closure is satisfied.

11.4 Test suite for cross-linguistic fieldwork

We propose a minimal elicitation suite that operationalizes the four central diagnostics cross-linguistically:

- (a) closure necessity (article/demonstrative/clitic agreement),
- (b) numeral specification,
- (c) partitive alternation, and
- (d) anchor sensitivity (remainder vs index).

Crucially, the unification claim predicts that remainder and index selectors will pattern together with respect to whichever closure device the language uses.

12. Processing and computational notes

This section sketches consequences for (i) psycholinguistics and neurolinguistics and (ii) computational linguistics and corpus annotation. The goal is not to overclaim existing experimental coverage, but to identify clear, testable predictions implied by the D-locked selector analysis.

12.1 Psycholinguistic and neurolinguistic expectations

If SHEs are D-locked selector constituents, the parser should treat ««the + X»» as a strong cue to a selector configuration. This predicts reduced ambiguity compared to ordinary adjective+noun sequences. Concretely, upon encountering «the other...», the processor can commit to a remainder-selection structure even before seeing a noun, whereas «the red...» typically invites a prediction of a following noun. Such differences

can be tested via self-paced reading, eye-tracking, and ERP paradigms that compare matched continuations (e.g., «the other two...» vs «the red two...»).

ERP studies have repeatedly associated semantic integration difficulty with N400 effects and syntactic reanalysis/repair with P600 effects (Kutas & Hillyard 1980; Osterhout & Holcomb 1992; Friederici 2011). The present account predicts that violating D-lock in selector readings («I took other») should trigger either (i) increased integration costs (N400-like) due to failed saturation of the selection variable, or (ii) repair/reanalysis (P600-like) if the processor attempts to reinterpret the string under a non-selector parse. Likewise, anchor violations («the next» without an available ordering anchor) are predicted to increase late costs due to discourse repair.

At a neurocognitive level, SHEs are attractive probes because they are compact and controllable: a single functional marker («the») plus a single selector head introduces a selection variable whose domain constraints can be manipulated experimentally (partition vs order). This enables experiments that separate definiteness closure effects from content-noun effects, and that test whether remainder and index selectors recruit partially distinct discourse networks.

12.2 Computational representation and parsing

Most NLP pipelines treat «other/next/last» as adjectival modifiers of a nominal head, and dependency schemes typically attach the determiner to the nominal head. However, in selector readings the minimal closed unit is «[the+X]»; therefore, a computationally faithful representation should preserve the selector constituent as a unit for downstream tasks (coreference, ellipsis resolution, quantifier scope, and discourse modeling).

We propose two minimal implementation options:

- (i) **Constituent-preserving parse**: represent «the other» as a SelP or DP subconstituent in phrase-structure parsing, with the numeral/NP attached as a specifier or complement.
- (ii) **Dependency augmentation**: keep standard dependencies but add a relation label (e.g., `selhead`) marking that X is a selector head, and treat the numeral as depending on X with a typed relation (`selspec`). This retains compatibility with UD while exposing the selector configuration.

Figure 4. Computational pipeline for detecting and interpreting SHE constructions

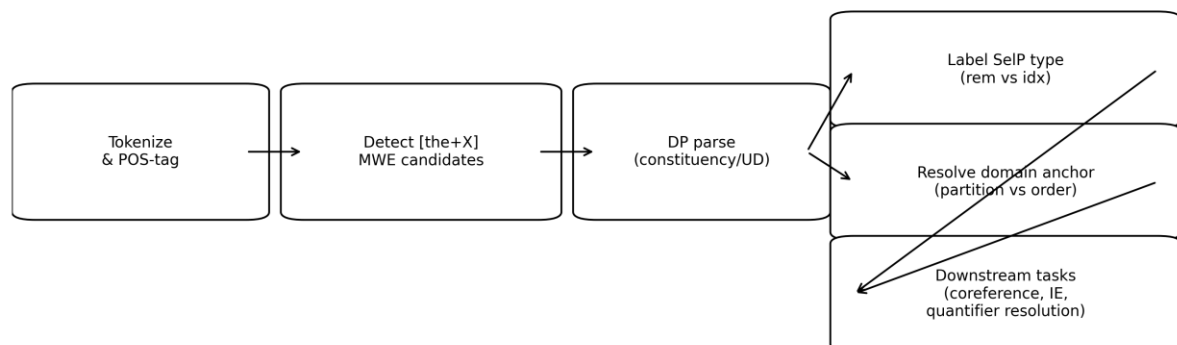


Figure. Computational detection pipeline for SHEs: candidate extraction, diagnostic scoring, and corpus-level validation.

12.3 Applications

Coreference and anaphora: SHEs often introduce discourse-linked complements or indices («the other», «the latter»), making them valuable triggers for coreference resolvers. Encoding the selector type (remainder vs index) can improve resolution by constraining candidate sets.

Information extraction: remainder selectors and index selectors encode set complements and list positions; extracting them explicitly benefits summarization and instruction understanding (e.g., procedural texts with «the next step», «the remaining items»).

Clinical and educational communication: because SHEs compactly encode selection over salient sets, they are frequent in instructions and narratives. A transparent model of their closure and anchoring can support controlled-language generation and reduce ambiguity in high-stakes communication.

13. Articles as closure operators beyond canonical noun heads

One of the broader implications of the SHE analysis is a reconceptualization of articles: English articles are not exclusively “privileged” for canonical noun heads. Rather, they can serve as *closure operators* that convert a range of non-nominal or operator-typed material into a DP-deployable term object. This section situates SHEs within a broader family of English constructions where an article closes a non-canonical head.

13.1 Substantivized adjectives and participles

Classic cases include substantivized adjectives and participles: «the rich», «the poor», «the accused», «the unknown». These have often been treated as nominal ellipsis (with an implicit noun such as «people») or as adjectival heads that project nominally under DP. Regardless of implementation, they show that D can close an adjectival head into a nominally usable unit (Huddleston & Pullum 2002).

13.2 Quantifier-like article closures

Another relevant family consists of quantifier-like closures such as «a few», «a little», and related measure expressions. Here the article appears to combine with an adjective-like item («few», «little») to yield a quantificational DP. These constructions further undermine the idea that articles are tied exclusively to canonical nouns. Instead, articles participate in creating closed term objects that can host in argument positions.

13.3 SHEs as a distinct closure family

SHEs extend this pattern in a theoretically significant way. Unlike substantivized adjectives (property-based) and quantifier-like closures (quantity-based), SHEs are **selection-based**: the closed head introduces a remainder or index variable. This is why the determiner is not merely licensing DP presence but is functionally required for saturating the selection variable. In this sense, SHEs provide a particularly clean case where the article is an operator over operator-typed material.

From the ULD–TBH perspective, these families are unified by a single idea: closure converts pre-nominal or operator-typed material into a one-term sign-hood entity capable of categorical hosting. SHEs provide a new data point supporting that general thesis.

14. Conclusion

This article proposed that English contains a small but robust DP-internal class—**Selector-Head Expressions** (SHEs) – that is not satisfactorily captured by standard labels such as adjective, determiner, or noun. The defining property is **D-lock**: in selector readings, the selector head introduces a discourse-anchored selection variable that cannot be saturated without DP closure, making «[the + X]» the minimal closed selector constituent.

Empirically, the proposal unifies two domains often treated separately: remainder selection («the other/rest/remaining/remainder») and indefinite ordinal/index selection («the next/following/previous/first/last/former/latter»). The unification is implemented via a typed Sel head (Sel_rem vs Sel_idx) under a single SelP constituency. A tiered diagnostic table and an explicit scoring protocol were provided so that candidate classification becomes reproducible rather than impressionistic.

Beyond English, the analysis predicts that languages without articles will realize the same closure requirement through alternative devices (demonstratives, clitics, agreement, classifiers, or discourse anchoring), and it yields clear predictions for processing and computational modeling. Finally, the SHE class supports a broader architectural claim of the monograph: articles function as closure operators beyond canonical noun heads, contributing directly to the creation of lexical-term objects (TBH) that can enter higher hosting regimes.

This appendix is an insert-only empirical module. It supplies (i) an acceptability table design with two high-leverage diagnostics that alternative parses struggle to mimic, (ii) corpus query patterns for validating numeral attachment to the closed selector constituent [the + X], and (iii) a cross-linguistic mini-dataset template aligned with the Sel_rem / Sel_idx typing.

D.1 Acceptability study design (pilot): D-locking and numeral attachment

Goal. Convert the constituency hypothesis into reviewer-visible quantitative evidence. The design targets the claim that [the + X] is the minimal closed selector constituent in

the selector reading, and that numerals/NPs attach as specifications of that closed unit (rather than composing with X alone).

D.1.1 Participants, task, and rating scale

- Participants: N = 20–40 native English speakers (pilot acceptable at N≈20; preregistered follow-up at N≥40).
- Task: acceptability rating (Likert 1–7) or forced-choice preference (A/B) with randomized presentation.
- Context control: each item is preceded by a short discourse prompt that forces the selector reading (set complement or ordered sequence).
- Design: within-subject; Latin-square counterbalancing across conditions to avoid repeated lexicalization in the same participant.

D.1.2 Conditions and predictions

Let X be a candidate selector head (e.g., other, rest, remaining, next, following, latter). Let NUM be an overt numeral (two/three or digits). The contrasts below are the core tests; they can be instantiated across the full inventory.

Condition	Form (schematic)	Example (selector context)	Prediction under D-lock	Primary diagnostic target
C1 (SHE-licensed)	[the + X] + NUM (+ N)	I used the other two tickets.	High	Baseline grammatical selector DP
C2 (D-omission)	X + NUM (+ N)	«I used other two tickets.	Low (selector reading)	Necessity of definite closure for selector saturation
C3 (standalone nominal)	[the + X] (Ø specification)	I used the other.	High	Constituency of [the + X] as a closed unit
C4 (specifier variation)	[the + X] + (two or three) (+ N)	I used the other two or three tickets.	High	NUMP attaches to [the+X], not to X alone
C5 (partitive option)	[the + X] + of + DP	I used the rest of the tickets.	High (where licensed)	Alternation between specification vs partitive PP

D.1.3 Two additional diagnostics hard for alternative parses to mimic

The two diagnostics below strengthen the D-locked constituency claim by exploiting asymmetries in ellipsis and coordination that follow naturally if [the + X] is a closed selector constituent. Each diagnostic is implemented as a controlled acceptability contrast with matched discourse prompts.

Diagnostic 1: Ellipsis asymmetry (closed-selector ellipsis)

Core idea. If [the + X] is closed, it should be licensed as a standalone nominal under ellipsis in contexts where the specification (NUM/NP) is recoverable.

1. Context: There were five tickets. Two were used already.
2. Target A (closed selector): I used the other two; you used the other three.
3. Target B (standalone selector): I used the other; you used the other three.
4. Negative control (bare X under ellipsis): «I used other; you used the other three.

Diagnostic 2: Targeted coordination with selector heads (SelP coordination)

Core idea. Coordination can test whether numeral specification attaches outside the selector constituent. If NUM/NP specifications attach to [the + X], parallel selector DPs should be preferred.

5. Context: There were ten chairs in a row; we are assigning them by position.
6. Target A (parallel selector DPs): I will take the next two, and you will take the last two.
7. Target B (selector-head coordination with shared specification; expected degraded): ?I will take the next and the last two.
8. Target C (bare X coordination; negative control): «I will take next and last two.

D.1.4 Materials worksheet (ready to use)

Item ID	X (candidate)	Selector type	Condition	Discourse prompt	Target sentence	Expected rating	Notes
R1-C1	other	Sel_rem	C1	There were five tickets; two were used.	I used the other two tickets.	High	Avoid idioms (other hand/side).
R1-C2	other	Sel_rem	C2	There were five tickets; two were used.	«I used other two tickets.	Low	Keep selector reading explicit.
I1-C1	next	Sel_idx	C1	There are ten chairs in a row; assign by	I will take the next two	High	Provide ordering anchor

				order.	chairs.		(row/list).
I1-C2	next	Sel_idx	C2	There are ten chairs in a row; assign by order.	«I will take next two chairs.	Low	Control non-selector uses by context.

D.2 Corpus validation: query patterns for numeral attachment to [the + X]

Goal. Provide a minimal corpus check that quantifies how often numerals follow [the + X] and compares these rates against near-alternatives.

D.2.1 Candidate set

- Sel_rem: other, rest, remaining, remainder.
- Sel_idx: next, following, subsequent, previous, prior, earlier, former, latter, first, last.

D.2.2 Regex-style patterns (tokenized text)

Pattern name	Regex (schematic)	Counts as	Notes
the X [0-9]+	\bthe\s+(X)\s+([0-9]+)\b	[the+X]+N UM	High precision
the X one two ... ten	\bthe\s+(X)\s+(one two three four five six seven eight nine ten)\b	[the+X]+N UM	Extend to teens/tens as needed
the rest of	\bthe\s+(rest remainder)\s+of\b	[the+X]+of	Key alternation for Sel_rem
X NUM (control)	\b(X)\s+(NUM)\b	X+NUM	Negative control (reading depends on context)

D.3 Cross-linguistic mini-dataset template

Goal. Provide a compact template that records the functional analogue of D-locking in languages with different DP-closure strategies (articles, demonstratives, agreement, clitics, classifiers).

Lang uage	Clos ure devi ce	Sele ctor type	Candi date X	Standa lone nomin al?	Num eral spec ?	Partit ive?	Exa mple	Gl oss	Transl ation	No tes	Sou rce

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Appendix A. Minimal test suite (English)

This appendix provides a compact set of minimal pairs that instantiate the diagnostics. Asterisks mark degraded strings in the selector reading; alternative non-selector readings may be possible in some contexts.

- A1 (D-lock): a. I'll take ««the other»». b. «I'll take ««other»».
- A2 (numeral specification): a. Take ««the next two»». b. Take ««the following two»». c. ?Take ««the subsequent two»».
- A3 (remainder vs index): a. We chose three; ««the rest»» were discarded. b. We chose three; ««the next»» were discarded. (requires list anchor)
- A4 (partitive): a. ««the rest of»» the group b. ««the remainder of»» the funds c. ?««the next of»» the group
- A5 (coordination): I'll take ««the other»» two and you take ««the next»» two.
- A6 (former/latter anchor): We discussed two plans. ««The former»» was rejected; ««the latter»» was revised.
- A7 (majority/minority): ««The majority»» voted yes; ««the minority»» voted no.
- A8 (negative control): a. ««the red two»» (≈ 'the two red ones') b. ««the other two»» (selector; remainder).
- A9 (ellipsis): A: Which pages? B: ««the next»» (ones).
- A10 (anchor violation): ??In an unordered pile, ««the next»» item is missing. (requires ordering anchor).

A11 (ordinal–numeral interaction): ««the first two»» chapters / ««the last three»» minutes.

A12 (remainder–numeral): ««the remaining two»» patients / ««the other ten»» items.

Appendix B. Extended candidate inventory (illustrative)

This appendix lists additional candidates that sometimes behave like SHEs but exhibit weaker diagnostic stability. These are not required for the core analysis, but they support the claim that SHE-like behavior extends to a broader zone of the lexicon.

Table B1. Illustrative peripheral candidates (subset).

Candidate	Standalone nominal	Numeral spec.	Partitive	Selector type	Tier
the prior	Yes (register-bound)	Limited	Marginal	Index	Peripheral
the earlier	Yes (register-bound)	Limited	Marginal	Index	Peripheral
the adjacent	Limited	Limited	No	Spatial subset	Peripheral
the nearby	Limited	Limited	No	Spatial subset	Peripheral
the opposite	Yes	Limited	No	Spatial subset	Peripheral

Appendix C. Implementation sketch for corpus detection

A practical detection pipeline can be implemented with three steps: (i) candidate extraction of «the + X» bigrams where X is in a candidate lexicon; (ii) scoring by distributional proxies (occurrence with numerals, standalone nominal usage via punctuation/ellipsis contexts, co-occurrence with «of»-PPs); and (iii) manual validation on a stratified sample. The classifier rubric in Section 9 provides a minimal set of features; computationally, these are implemented as counts over dependency patterns and local POS contexts.

For applications that require explicit representation of SelP, phrase-structure parsers can be adapted with a small grammar fragment that introduces SelP between D and NumP/NP. Alternatively, dependency representations can add a typed feature `SelectorHead=Yes` on X and preserve standard edges. Both strategies enable downstream reasoning about remainder and index selection.

Appendix D. Empirical modules for stress-testing the SHE constituency claim